

MATH 4340/ COSC 4340 – Numerical Methods for Differential Equations

HW 4: Due by 2:45pm on Tuesday February 20, 2024.

Submit to WyoCourses a zipped folder containing a Jupyter notebook (both ipynb and HTML files).

Handwritten work should be included as a PDF or typed into the notebook as LaTeX.

1. [20 points] For each of the following 2-step methods, state whether its implicit vs explicit, find whether it is stable, and find the order of accuracy:

$$(A) \quad y_{i+1} = y_i + \frac{h}{4}[7f(t_i, y_i) - 3f(t_{i-1}, y_{i-1})]$$

$$(B) \quad y_{i+1} = 3y_i - 2y_{i-1} + \frac{h}{3}[2f(t_{i+1}, y_{i+1}) + f(t_i, y_i) - 6f(t_{i-1}, y_{i-1})]$$

2. [30 points] Consider the 2-step implicit Adams-Moulton method

$$y_{i+1} = y_i + \frac{h}{12}[5f(t_{i+1}, y_{i+1}) + 8f(t_i, y_i) - f(t_{i-1}, y_{i-1})].$$

- (A). [10 points] Use the Dahlquist Theorem to argue that the method is convergent.
- (B). [10 points] Let $f(t, y) = \lambda y$ with $\lambda = -2$ and solve the IVP using this method with $N = 100$, $[t_0, T] = [0, 5]$, and initial conditions $y(t_0) = 2$ and $y(t_1) = 2e^{\lambda t_1}$. (Note that $t_1 = t_0 + (T - t_0)/N = 0.05$.) Make a plot comparing the approximation to the true solution $y(t) = 2e^{\lambda t}$, and make a second plot for the errors E_i .
- (C). [10 points] Repeat Problem 2B two times using two different choices for λ : $\lambda = -20$ and $\lambda = -200$. What happens as λ becomes really negative? Why? *Hint: Recall regions of absolute stability.*

3. [50 points] Consider the implicit trapezoidal method

$$y_{i+1} = y_i + \frac{h}{2}[f(t_{i+1}, y_{i+1}) + f(t_i, y_i)].$$

- (A). [10 points] Let $N = 2000$, $[t_0, T] = [0, 1]$ and $y(0) = 0.5$ and solve the following stiff ODE using both the explicit and implicit trapezoidal methods.

$$\frac{d}{dt}y = f(t, y) = 50 * (1 - y).$$

Create a plot showing both approximations and a second plot showing the global errors E_i for both methods, which can be computing using the true solution $y(t) = 1 - e^{-50t}$.

- (B). [10 points] Repeat Problem 3A two times with two different choices of N : first $N = 200$ and then $N = 20$. Describe what you observe using knowledge about the utility of implicit and explicit methods for stiff equations.
- (C). [15 points] Observe that $y = 1$ is an attracting fixed point and under the change of variables $x(t) = 1 - y(t)$ one finds an equivalent system $\frac{dx}{dt} = -50x$. Prove that the explicit trapezoidal method for this equation is identical to the fixed point iteration $x_{i+1} = g(x_i) = (1 - 50h + 1250h^2)x_i$. Then prove that this iteration does not converge when $N < 25$ (i.e., h is too large and the method is not “absolutely stable”).
- (D). [5 points] Repeat Problem 3B, but now choose $N = 26$ and then $N = 24$. What happens as N decreases below 25?
- (E). [10 points] Again consider the equivalent system $\frac{dx}{dt} = \lambda x$ with $\lambda = -50$, and prove that the implicit trapezoidal method is absolutely stable for all h . (Recall that methods that are absolutely stable for all $h\lambda < 0$ are called ‘A stable.’)